An Exact Event-Time Atlas for an Affine-Impact Bouncing Ball

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Abstract

We analyse a gravity-driven ball on a half-line with restitution and a constant impact impulse. The physical flow is $\dot{q} = v$, $\dot{v} = -g$, the guard is $q = 0, v^- < 0$, and the reset is $v^+ = r(-v^-) + J$. On the positive outgoing section the event map is the affine map $P(u) = ru + J$, but each iterate carries the physical roof $2u/g$. This gives exact geometric and arithmetic event sequences, a sharp Zeno boundary, a unique positive forced flight for $J > 0, r < 1$, and separate elastic, sticking, and accelerating boundaries. We adjoin $(0, 0)$ as rest and exclude it from the regular section, so a zero-duration fixed event is not counted as a physical cycle. The certificate checks 96 exact event-time cells and 36 interior impact reconstructions. Any fixed-point series is explicitly an event-map object, not a physical-flow zeta or a target arithmetic construction.

1 Hybrid model and reduction

For $g > 0$, $0 \le r \le 1$, and $J \ge 0$, let

$$\dot{q} = v, \qquad \dot{v} = -g \quad (q > 0), \qquad v^+ = r(-v^-) + J \quad (q = 0, v^- < 0).$$

The point $(q, v) = (0, 0)$ is a separate absorbing rest state. For an interior state, solving $q_0 + v_0 t - gt^2/2 = 0$ gives

$$\tau_0 = \frac{v_0 + \sqrt{v_0^2 + 2gq_0}}{g}, \qquad w_0 = \sqrt{v_0^2 + 2gq_0}, \qquad u_0 = rw_0 + J.$$

Starting just after an impact with positive outgoing speed u, symmetry of constant acceleration gives one physical flight of duration $2u/g$ and returns the incoming speed $-u$. Thus the outgoing section map is

$$P(u) = ru + J, \qquad \tau(u) = \frac{2u}{g}, \qquad u \in \mathcal{S}_+ = (0, \infty).$$

Theorem 1 (event-time atlas)**.** *For $r \neq 1$, put $u_* = J/(1 - r)$. Then*

$$u_n = u_* + r^n(u_0 - u_*), \qquad t_n = \frac{2}{g}\left[nu_* + (u_0 - u_*)\frac{1 - r^n}{1 - r}\right].$$

For $r = 1$, $u_n = u_0 + nJ$ and $t_n = \frac{2}{g}[nu_0 + Jn(n - 1)/2]$. If $J = 0$ and $0 < r < 1$, positive speeds form a Zeno sequence with accumulation time $2u_0/[g(1 - r)]$. The edge $r = 0, J = 0$ has one positive-duration flight at most and then sticks at rest. If $J > 0$ and $r < 1$, u_* is the unique positive forced cycle, with physical period $2u_*/g$ and event multiplier $P'(u) = r$; every positive speed converges to it. At $r = 1, J = 0$ there is a continuum of elastic periods $2u/g$, while $r = 1, J > 0$ is nonperiodic with quadratic event times.

Proof. The first equation is the quadratic flight root. The reset gives P, and induction followed by a finite geometric or arithmetic sum gives the displayed iterates and times. The geometric sum is finite precisely for $J = 0, 0 < r < 1$; $r = 0$ reaches zero after one flight and the rest convention stops the execution. Solving $u = P(u)$ and using contraction proves the forced cycle and its multiplier. Identity and translation follow directly. □

2 Section domains and formal series

The regular section is $\mathcal{S}_+ = (0, \infty)$ (the literal label `S_+`): it parametrizes positive-duration flights only. Consequently, for $J > 0, r < 1$ the physical event-map fixed-point series is $\zeta_{\text{phys}}(z) = 1/(1-z)$, whereas for $J = 0, r < 1$ it is the empty series 1 on $\mathcal{S}_+$. If the section is artificially closed to $[0, \infty)$, the affine boundary point $u = 0$ contributes the separate formal series $\zeta_{\text{aff}}(z) = 1/(1-z)$ when $J = 0, r < 1$; that point is rest, not a physical orbit. For $r = 1, J = 0$ the fixed set is a continuum and a cardinality series is undefined. For $r = 1, J > 0$ translation has no fixed point, so its finite-count series is 1. None of these event-map series is a physical-flow zeta.

3 Exact audit and scope boundary

Twelve rational controls cover contraction, strict Zeno, the $r = 0$ sticking edge, elastic identity, and translation. The ledger contains 96 exact roof and cumulative-time cells and 36 first-impact reconstructions. An independent checker passes 368 exact assertions; SymPy passes 11 identities, byte replay is exact, and 14 repaired/stale hash mutations are rejected. In particular, an attack that relabels $r = 0, J = 0$ as Zeno is caught before release.

The Route-A tuple is

```
(A0_FAIL, A1_WEAK, A2_FAIL, A3_FAIL, A4_FORMAL_HINT),
```

with `overall=ROUTE_A_REJECTED` and `route_b_invocation_allowed=false`. The package makes no target prime, zero, local factor, root number, automorphy, target functional equation, or Hilbert–Polya operator claim.

Source note

Hybrid modeling context is given by Leine–Nijmeijer [1] and Goebel–Sanfelice–Teel [2]; no priority claim is made.

References

[1] R. I. Leine and H. Nijmeijer, *Dynamics and Bifurcations of Non-Smooth Mechanical Systems*, Springer (2004), DOI:10.1007/978-3-540-44398-8.

[2] R. Goebel, R. G. Sanfelice, and A. R. Teel, *Hybrid Dynamical Systems: Modeling, Stability, and Robustness*, Princeton University Press (2012), DOI:10.1515/9781400842636.

Declarations

Scope. The exact registered literal is `NO_BAD_EULER_OR_ROOT_NUMBER`. **Data and code.** The theorem, exact ledger, independent checks, and build recipe are released with HCS-C212. **Competing interests.** None declared. **AI-use disclosure.** Generative tools assisted drafting and code generation; the internal artifact chain checked formulas, metadata, and scope boundaries. This is not external peer review.